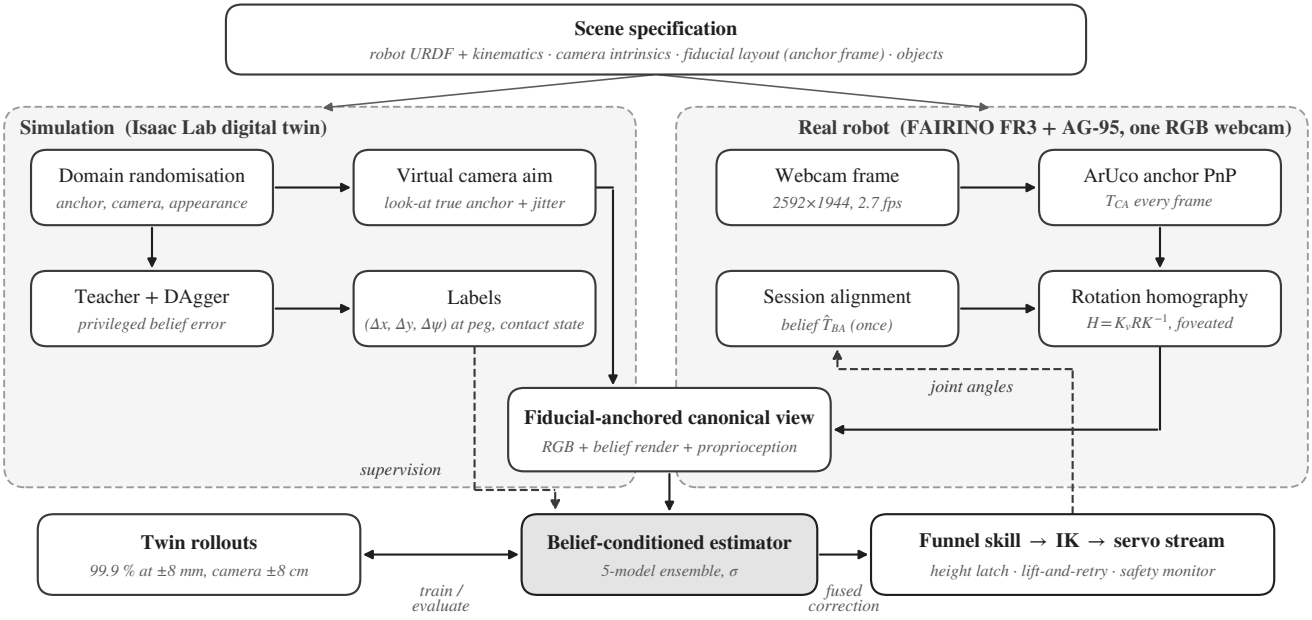
Scene specification
robot URDF + kinematics · camera intrinsics · fiducial layout (anchor frame) · objects
Simulation (Isaac Lab digital twin)
Domain randomisation
anchor, camera, appearance
Virtual camera aim
look-at true anchor + jitter
Teacher + DAgger
privileged belief error
Labels
(Δx, Δy, Δψ) at peg, contact state
supervision
Real robot (FAIRINO FR3 + AG-95, one RGB webcam)
Webcam frame
2592×1944, 2.7 fps
ArUco anchor PnP
T_{CA} every frame
Session alignment
belief $\hat{T}_{BA}$ (once)
Rotation homography
$H = K_v R K^{-1}$, foveated
joint angles
Fiducial-anchored canonical view
RGB + belief render + proprioception
Twin rollouts
99.9 % at ±8 mm, camera ±8 cm
train / evaluate
Belief-conditioned estimator
5-model ensemble, σ
fused correction
Funnel skill → IK → servo stream
height latch · lift-and-retry · safety monitor